XRPL Validator — Daily Engineering Report · 2026-11-06

Period: 2026-11-06 00:00 to 24:00 US Eastern (2026-11-06 05:00 to 2026-11-07 05:00 UTC); ledgers #108,146,084 to #108,167,677. Every count from the ledger and the validator's logs stops at #108,167,677, closed 04:59:51 UTC. Generated 2026-11-07 00:20 EST by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Degraded — 2 engine receipts open.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,592 of the 21,594 ledgers the network closed (99.991%); every validation matched the network's validated ledger. Unsigned: 0 during planned restarts, 2 outside maintenance (0 incidents, 2 isolated misses).
- **Native engine:** 2 engine receipts: #108,167,573 — open; #108,167,575 — open.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Network:** 7 slow-close periods; the longest, 14:40 UTC–14:42 UTC, closed ledgers up to 58 s apart.
- **Independence program:** 9 of 30 qualifying days (snapshot at 05:21:05 UTC).

Service levels

Measure	Value
Ledgers closed by the network	21,594
Validations issued	21,592
Coverage (validations / ledgers closed)	99.991%
Validations matching the network's validated ledger hash	21,592 of 21,592
Unsigned during planned restarts	0
Unsigned outside maintenance	2
Coverage outside planned maintenance (objective 99.9%)	99.991%
Restarts	0
Basis: our own ledger hash	21,591
Basis: our own hash, witnessed by the reference node (quorum short)	1
External agreement, data.xrpl.org (UTC day 2026-11-06)	0.99990 (2 missed of 19,778)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#108,146,084–#108,167,677), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	2
Step 4 · peer data	0
Step 4 · validations	1
Step 5 · own hash	0
Writer	2

Live counters, snapshot at 05:21:05 UTC (since the last restart, 2026-11-05 00:49 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	46,514 ledgers, 10,757,731 transactions	46,513 fully matched; results 0, bytes 3, keys 0/0
2 · transaction records	46,993 ledgers, 10,897,141 records	10,897,104 exact; differ 37, tree 1
3 · transaction order	47,036 ledgers (mode: on)	47,033 exact; order 3, results 0, bytes 0, refused 3
4 · ledger data from peers	47,042 ledgers fetched	0 matched; differ 0, failed 0
4 · trusted validations	47,036 ledgers judged	47,018 agreed; differ 0, short of quorum 18
5 · our own ledger hash	47,040 ledgers	46,989 own; differ 1, partial 50
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	14:40 UTC	#108,154,884	Step 4 · validations	25 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)
R-02	04:53 UTC	#108,167,573	Step 3 · order, Writer	our transaction order differed at position 78	open
R-03	04:53 UTC	#108,167,575	Step 3 · order, Writer	our transaction order differed at position 5	open

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-11-06-01, INC-01 is INC-2026-11-06-01.

Incidents

No incidents.

Isolated unsigned ledgers (shorter than 3 ledgers, cause not recorded): #108,167,573 at 04:53:11 UTC, #108,167,575 at 04:53:20 UTC.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **9 of 30** (snapshot at 05:21:05 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-11-06 — tracked #108,141,635–#108,163,173	Value
Validations	21,539 (21,539 own, 0 borrowed)
Coverage of the tracked ledgers	100.00%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,594 (#108,146,084–#108,167,677)
Mean close interval	4.00 s
Longest close interval	58 s (before #108,154,885)
Closes at least 15 s apart	9

Measure	Value
Empty ledgers	0

- Slow closes 07:24:00 UTC–07:24:40 UTC: ledgers #108,148,260–#108,148,261 (2), 2 closes at least 15 s apart, up to 20 s.
- Slow closes 13:41:42 UTC–13:42:00 UTC: ledgers #108,154,010, 1 close at least 15 s apart, up to 18 s.
- Slow closes 14:32:50 UTC–14:33:10 UTC: ledgers #108,154,779, 1 close at least 15 s apart, up to 20 s.
- Slow closes 14:40:42 UTC–14:42:00 UTC: ledgers #108,154,885–#108,154,887 (3), 2 closes at least 15 s apart, up to 58 s.
- Slow closes 16:54:01 UTC–16:54:20 UTC: ledgers #108,156,821, 1 close at least 15 s apart, up to 19 s.
- Slow closes 16:54:50 UTC–16:55:10 UTC: ledgers #108,156,826, 1 close at least 15 s apart, up to 20 s.
- Slow closes 22:27:31 UTC–22:27:50 UTC: ledgers #108,161,802, 1 close at least 15 s apart, up to 19 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

No amendment held a majority.

Amendments enabled on mainnet: 98.

Open items

- Receipt R-02 (#108,167,573): open, under investigation.
- Receipt R-03 (#108,167,575): open, under investigation.
- Unsigned #108,167,573 (04:53:11 UTC): cause not recorded.
- Unsigned #108,167,575 (04:53:20 UTC): cause not recorded.

Corrections to earlier reports

None.